Direct Numerical Simulations of Mixed-Phase Cloud Glaciation Caused by Turbulent Mixing

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Key Points:

- For the same initial ice mass, glaciation is enhanced if ice crystals are smaller but many compared to larger but few.
- A higher level of flow turbulence increases the rate of glaciation for any ice number density.
- Compared to spherical ice crystals, glaciation is enhanced if ice crystal geometry is hexagonal column but is suppressed if the shape is hexagonal plate.
- The cloud droplet radius PDFs are bimodal and those of ice crystals are skewed to the left if the turbulence intensity is low.

Abstract

Glaciation of turbulent mixed-phase clouds is examined at the microscale using direct numerical simulations (DNS) and Lagrangian particle tracking. The effects of flow turbulence intensity, ice number density, and ice crystal geometry on glaciation are studied. The results suggest that high turbulence has an expediting effect on ice growth and glaciation. For the same initial ice mass, smaller but many ice crystals promote glaciation relative to larger but few ice crystals. Compared to spherical ice crystals, glaciation is enhanced if ice crystal geometry is hexagonal column but is suppressed if the shape is hexagonal plate. Neither ice number density nor ice geometry has any impact on cloud droplet evaporation. Turbulence mixing reduces the relative dispersions of cloud droplet and ice crystal radii. It is found that the size distributions of cloud droplets are bimodal and those of ice crystals are skewed to the left if the turbulence intensity is low.

1 Introduction

The saturation vapor pressure with respect to liquid water is higher than that with respect to ice. If the environmental supersaturation level is such that it is supersaturated with respect to ice but subsaturated with respect to water, ice crystals will grow while cloud droplets will shrink. If this continues, a mixed-phase cloud will turn into an ice cloud or glaciate (Korolev et al., 2003). This is the Wegener-Bergeron-Findeisen (WBF) process (Storelvmo & Tan, 2015), also commonly known as glaciation. But mixed-phase clouds have been found to be stable (Morrison et al., 2012), meaning that they do not fully glaciate and remain in a mixed phase condition. To understand the glaciation and persistence of mixed-phase clouds, the interactions between ambient turbulence, cloud droplets, and ice crystals need to be studied at the microscale.

The WBF mechanism was studied analytically by Korolev (2007) who showed that water vapor mass can increase or decrease during this process. An increase of water vapor means that evaporation of cloud droplets is faster than deposition of ice crystals while a decrease of water vapor indicates the opposite. The analytical investigation by Pinksly et al. (2014) suggested that ice crystal number density and initial liquid water mass strongly influence the rate of glaciation. Mixed-phase clouds had been studied numerically. Hill et al. (2013) studied the growth of cloud and ice using a large eddy simulation (LES) model. The authors reported that liquid water mass in a mixed-phase cloud decreases if the ice number density and ice crystal radius increase. Yang et al. (2015) applied LES to examine the growth of ice crystals in a mixed-phase cloud and found that a fraction of ice crystals get trapped in large eddies and grow much bigger than the rest. In their LES investigation, Chen et al. (2024) studied the evolution of mixed-phase clouds using direct numerical simulations (DNS) to report that the size distributions of cloud droplets broaden more compared to those of ice crystals. They further suggested that initial liquid mass should be large enough for mixed-phase condition to persist. Sarnitsky et al. (2024) used DNS to investigate mixed-phase cloud and reported that flow turbulence affects ice crystal growth and thus glaciation.

The geometry of ice crystals is important, particularly at the microscale. Using a theoretical approach, Chen and Lamb (1994) showed that the aspect ratio ice crystals changes as ice mass grows with time. Bailey and Hallett (2003) experimentally demonstrated that the complexity of ice crystal shape increases if the supersaturation with respect to ice is higher. Sheridan et al. (2009) suggested that growth is more isotropic if ice crystals are larger initially but more twisted if they are smaller. Sulia and Harrington (2011) found that the glaciation rate can be under-predicted by nearly an order of magnitude if equivalent spheres are assumed in place of the precise ice crystal shape. Hence, the role of ice crystal shape during glaciation needs improved understanding.

In the present study, we implement DNS with Lagrangian tracking of cloud droplets and ice crystals to study glaciation of mixed-phase clouds. For this purpose, we modify our previous DNS model (Sharfuddin et al., 2025) by including the ice phase equations. The cloudy and clear-air regions are initially separated by a sharp interface and subsequent turbulent mixing is simulated. We vary the intensity of flow turbulence and number density of ice crystals to assess their impacts on glaciation. The growth profiles of different ice crystal geometries: sphere, hexagonal column, and hexagonal plate, are also compared.

2 Governing Equations

The turbulent flow of air is governed by the incompressible Navier-Stokes equations:

$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u}.\nabla)\boldsymbol{u} = -\frac{1}{\rho_a}\nabla p + \nu\nabla^2\boldsymbol{u} + \boldsymbol{F}_c \,, \tag{1}$$

$$\nabla.\boldsymbol{u} = 0\,. \tag{2}$$

Here, $\boldsymbol{u}$ is the instantaneous velocity vector, ν is the kinematic viscosity, p is the instantaneous pressure and ρ_a is the density of air. $\boldsymbol{F}_c$ is an external force to sustain turbulence as described in Sharfuddin et al. (2025). The transport equations for the thermodynamic fields of temperature (T) and water vapor mixing ratio (q_v) are:

$$\frac{\partial T}{\partial t} + (\boldsymbol{u}.\nabla)T = \frac{L_w}{c_p}C_w + \frac{L_{ice}}{c_p}C_{ice} + \mu_T\nabla^2 T\,, \tag{3}$$

$$\frac{\partial q_v}{\partial t} + (\boldsymbol{u}.\nabla)q_v = -(C_w + C_{ice}) + \mu_v\nabla^2 q_v\,. \tag{4}$$

Above, L_w and L_{ice} are specific latent heats of condensation and deposition, respectively. C_w is the time rate of mass exchange between vapor and liquid while C_{ice} denotes the same between vapor and ice. Also, μ_T and μ_v are molecular diffusivities of temperature and vapor, respectively. The environmental supersaturation (S_e) depends on the saturation vapor mixing ratio ($q_{v,s}$) and the saturation vapor pressure (p_{sat}).

$$S_e = \frac{q_v}{q_{v,s}} - 1 = \frac{q_v}{621.97\,\frac{p_{sat}}{p - p_{sat}}} - 1\,, \tag{5}$$

p_{sat} can be calculated with respect to water and ice (Huang, 2018).

$$p_{sat} = \begin{cases} p_{sat,w} = 611.2 \times \exp\left(\dfrac{17.67(T - 273.15)}{T - 29.65}\right) \\ p_{sat,ice} = 611.2 \times \exp\left(\dfrac{22.46(T - 273.15)}{T - 0.53}\right) \end{cases}. \tag{6}$$

The cloud droplets and ice crystals are tracked individually in Lagrangian fashion. The equations for the position ($\boldsymbol{X}$) and velocity ($\boldsymbol{V}$) vectors can be written as:

$$\frac{d\boldsymbol{X}_i(t)}{dt} = \boldsymbol{V}_i(t)\,, \quad \frac{d\boldsymbol{V}_i(t)}{dt} = \frac{1}{(\tau_p)_i}\left[\boldsymbol{u}_i(\boldsymbol{X},t) - \boldsymbol{V}_i(t)\right] + \boldsymbol{g}\,, \quad (\tau_p)_i = \frac{2\rho_i (r_i)^2}{9\rho_a \nu}\,. \tag{7}$$

where the subscript 'i' denotes the i-th particle (cloud droplet or ice crystal). $\boldsymbol{u}_i$ is the instantaneous fluid velocity vector interpolated at the position of particle i and τ_p is the particle response time. Conveniently, the S_e and particle radius (r) can be written as:

$$S_e = \begin{cases} S_{e,w} & if \quad p_{sat} = p_{sat,w} \\ S_{e,ice} & if \quad p_{sat} = p_{sat,ice} \end{cases}, \quad r = \begin{cases} r_w & if \quad S_e = S_{e,w} \\ r_{ice} & if \quad S_e = S_{e,ice} \end{cases}. \tag{8}$$

The growth equations for cloud droplets and ice crystals can be written as:

$$\frac{d\big(r_w(t)\big)_i}{dt} = \left(\frac{G_w}{r_w(t)} S_{e,w}(\boldsymbol{X}, t)\right)_i , \qquad \frac{d\big(r_{ice}(t)\big)_i}{dt} = \left(\frac{G_{ice}}{r_{ice}(t)} S_{e,ice}(\boldsymbol{X}, t)\right)_i . \tag{9}$$

G_w and G_{ice} are expressed as:

$$G_w = \frac{1}{\frac{L_w \rho_w}{k_T T}\left(\frac{L_w}{R_v T} - 1\right) + \frac{\rho_w R_v T}{\mu_v p_{sat,w}}} , \qquad G_{ice} = \frac{1}{\frac{L_{ice} \rho_{ice}}{k_T T}\left(\frac{L_{ice}}{R_v T} - 1\right) + \frac{\rho_{ice} R_v T}{\mu_v f p_{sat,ice}}} . \tag{10}$$

where k_T is the thermal conductivity in air, R_v is the specific gas constant for water vapor, and f is the ice crystal shape factor. The f is unity for spherical ice crystals. For hexagonal ice crystals, the f is found as: (Westbrook et al., 2008)

$$f = 0.58(1 + 0.95\mathrm{A}^{0.75})(b/r_{ice}) , \qquad \mathrm{A} = \frac{h}{2b} . \tag{11}$$

Above, b is the side length and h is the height of a regular hexagonal prism, and A is the aspect ratio. The shape is hexagonal column if $\mathrm{A} > 1$ and hexagonal plate if $\mathrm{A} < 1$. r_{ice} is the equivalent radius of hexagonal ice crystals and is related to b and h as:

$$\frac{4}{3}\pi r_{ice}^3 = \frac{3\sqrt{3}}{2} b^2 h , \quad r_{ice} = 0.8527(b^2 h)^{1/3} , \quad b = 0.93\left(\frac{r_{ice}}{\mathrm{A}^{1/3}}\right) . \tag{12}$$

We neglect processes like ice nucleation, secondary ice production, branching, aggregation, and ventilation in our model and remain focused on diffusion-limited ice growth. With the foregoing, C_w and C_{ice} can be determined as:

$$C_w(\boldsymbol{x}, t) = \frac{4\pi\rho_w}{\rho_a a^3} \sum_{i=1}^{n_w} \Big(G_w r_w(t) S_{e,w}(\boldsymbol{X}, t)\Big)_i , \tag{13}$$

$$C_{ice}(\boldsymbol{x}, t) = \frac{4\pi\rho_{ice}}{\rho_a a^3} \sum_{i=1}^{n_{ice}} \Big(G_{ice} r_{ice}(t) S_{e,ice}(\boldsymbol{X}, t)\Big)_i . \tag{14}$$

where ρ_w and ρ_{ice} are liquid and ice densities, respectively. n_w is the number of cloud droplets in a grid cell with a volume of a^3 and n_{ice} is the same for ice crystals.

3 Numerical Details

To vary the flow turbulence level, we force different wavenumber bands as discussed in Sharfuddin et al. (2025). Specifically, we use the wavenumber bands: $1 \leq k_f/k_{min} \leq 3.0$ and $1 \leq k_f/k_{min} \leq 16.18$ to generate ‘low’ and ‘high’ turbulence levels and denote them as ‘L’ and ‘H’, respectively. k_f is the forced wavenumber and k_{min} is the minimum wavenumber. The dissipation rates of turbulent kinetic energy (TKE) are 0.01388 m^2/s^3 and 0.00037 m^2/s^3 and the Kolmogorov length scales are 0.00092 m and 0.00226 m for ‘low’ and ‘high’ turbulence levels, respectively.

The physical model simulated is that of a cubic domain inside which turbulence, cloud droplets, and ice crystals interact. The domain volume is 0.512^3 m^3 and the boundary conditions are triply periodic. At the start of simulations, the domain is divided equally into cloudy and clear-air regions. Cloud droplets and ice crystals are placed in the cloudy region only and subsequent turbulent mixing causes them to spread to all over the domain. A slab-like setup is assumed for the initial field to enable a sharp transition in the vapor mixing ratio and temperature at the cloud/clear-air interface (Sharfuddin et al., 2025). The initial vapor mixing ratio field is:

$$q_v(z,t=0)=\begin{cases} q_v^{max}, & d-L\times 0.5\times 0.5\le z<d+L\times 0.5\times 0.5 \\ q_{v,e}, & elsewhere \end{cases}. \quad (15)$$

Above, $q_v^{max}=2.367\ g/kg$, $q_{v,e}=1.90\ g/kg$, and $d\equiv L/2$ is the width of the cloud slab. q_v^{max} corresponds to the cloudy region and $q_{v,e}$ represents the dry (subsaturated) clear-air region. The temperature field is initialized as:

$$T(t=0)=\langle T(t=0)\rangle-0.608\langle T(t=0)\rangle(q_v(z,t=0)-\langle q_v(t=0)\rangle)\,. \quad (16)$$

We simulate eight cases: L-H, H-H, L-L, H-L, L-H-C, L-H-P, H-H-C, and H-H-P, for which the details are shown in Table I. Table I also lists the values of relevant parameters. Note that higher ice number density corresponds to smaller mean ice crystal radius and the vice versa in our model because initial ice masses are equal for all cases. Ice crystals are hexagonal column for Cases L-H-C and H-H-C, hexagonal plate for Cases L-H-P and H-H-P, and spherical otherwise. For hexagonal column, the values of f, b, and h are 1.148, 5.14 μm, and 25.69 μm, respectively, at the initial time. The corresponding values are 1.082, 9.46 μm, and 7.57 μm for hexagonal plate. The cloud droplet number density is 59.6 cm^{-3} for all the cases and is much higher than that of ice crystals, which is typical for mixed-phase clouds. The number of grid points is 256^3 and the grid resolution is 2 mm for all eight cases.

Our setup differs from that of Chen et al. (2024) because we assume an inhomogeneous distribution in the initial scalar fields with values changing sharply at the interface (Equations (15) and (16)). This allows us to realistically simulate turbulent mixing and evolution of the mean and fluctuating thermodynamic properties. Our DNS model uses a fifth order finite difference scheme to solve the convective terms in Equations (1), (3), and (4) as sharp interfaces are present. Whereas Chen et al. (2024) used a pseudospectral scheme.

Table 1

Details of the test cases

Case	TKE dissipation rate (m^2/s^3)	Ice number intensity (cm^{-3})	Initial ice crystal radius (μm)	Initial cloud droplet radius (μm)	Aspect ratio (h/b) of ice crystals
L-H	0.00037	0.011	7.5	7.5	N/A
H-H	0.01388	0.011	7.5	7.5	N/A
L-L	0.00037	0.0011	16.16	7.5	N/A
H-L	0.01388	0.0011	16.16	7.5	N/A
L-H-C	0.00037	0.011	7.5	7.5	2.5
L-H-P	0.00037	0.011	7.5	7.5	0.4
H-H-C	0.01388	0.011	7.5	7.5	2.5
H-H-P	0.01388	0.011	7.5	7.5	0.4

Case	Comments
L-H	low (L) turbulence, high (H) ice number density, spherical
H-H	high (H) turbulence, high (H) ice number density, spherical

L-L	low (L) turbulence, low (L) ice number density, spherical
H-L	high (H) turbulence, low (L) ice number density, spherical
L-H-C	low (L) turbulence, high (H) ice number density, hexagonal column (C)
L-H-P	low (L) turbulence, high (H) ice number density, hexagonal plate (P)
H-H-C	high (L) turbulence, high (H) ice number density, hexagonal column (C)
H-H-P	low (L) turbulence, high (H) ice number density, hexagonal plate (P)

Parameters and their values

Symbol	Value	Unit	Symbol	Value	Unit
L_w	2.50×10^6	J/kg	c_p	1005.0	J/kg/K
L_{ice}	2.83×10^6	J/kg	ρ_a	0.7674	kg/m^3
ρ_w	1000	kg/m^3	ρ_{ice}	917	kg/m^3
$p(t=0)$	57160	N/m^2	$\langle T(t=0) \rangle$	259.53	K
R_v	461.5	J/kg/K	k_T	0.0238	W/m/K
μ	1.64×10^{-5}	m^2/s	$\mu_T = \mu_v$	2.2×10^{-5}	m^2/s

4 Results and Discussion

The mean supersaturation is negative with respect to water but positive with respect to ice at the initial time. Hence, ice crystals grow but cloud droplets evaporate, and glaciation takes place. We vary the flow turbulence level, ice number density, and ice crystal aspect ratio. Their effects on thermodynamics and microphysics are discussed.

Figures 1(a)-(d) show the temporal evolution of mean thermodynamic fields. Liquid water converts into water vapor during evaporation of cloud droplets and ice crystals grow by absorbing water vapor. So, water vapor increases and decreases at the same time but there is a net increase (Figure 1(b)). This is because evaporation is dominant over ice deposition as cloud droplet mass exceeds that of ice crystals. The increase of $\langle q_v \rangle$ is accompanied by a decrease of $\langle T \rangle$ because more energy is absorbed through evaporation than released through deposition. The $\langle S_{e,w} \rangle$ and $\langle S_{e,ice} \rangle$ are directly proportional to the $\langle q_v \rangle$ and increase similarly (Figures 1(c)-(d)). The increase of $\langle q_v \rangle$ is faster for high turbulence cases (H-H and H-L) compared to low turbulence cases (L-H and L-L). Turbulence accelerates cloud droplet evaporation and produces more water vapor by enhancing the mixing between dry air and cloud. It is evident that ice number density or ice crystal radius has no impact on the mean thermodynamic fields which are identical for Cases H-H and H-L and for Cases L-H and L-L. We also find that the mean profiles in Figures 1(a)-(d) change more slowly at late times because the system becomes more homogeneous over time due to continuous mixing.

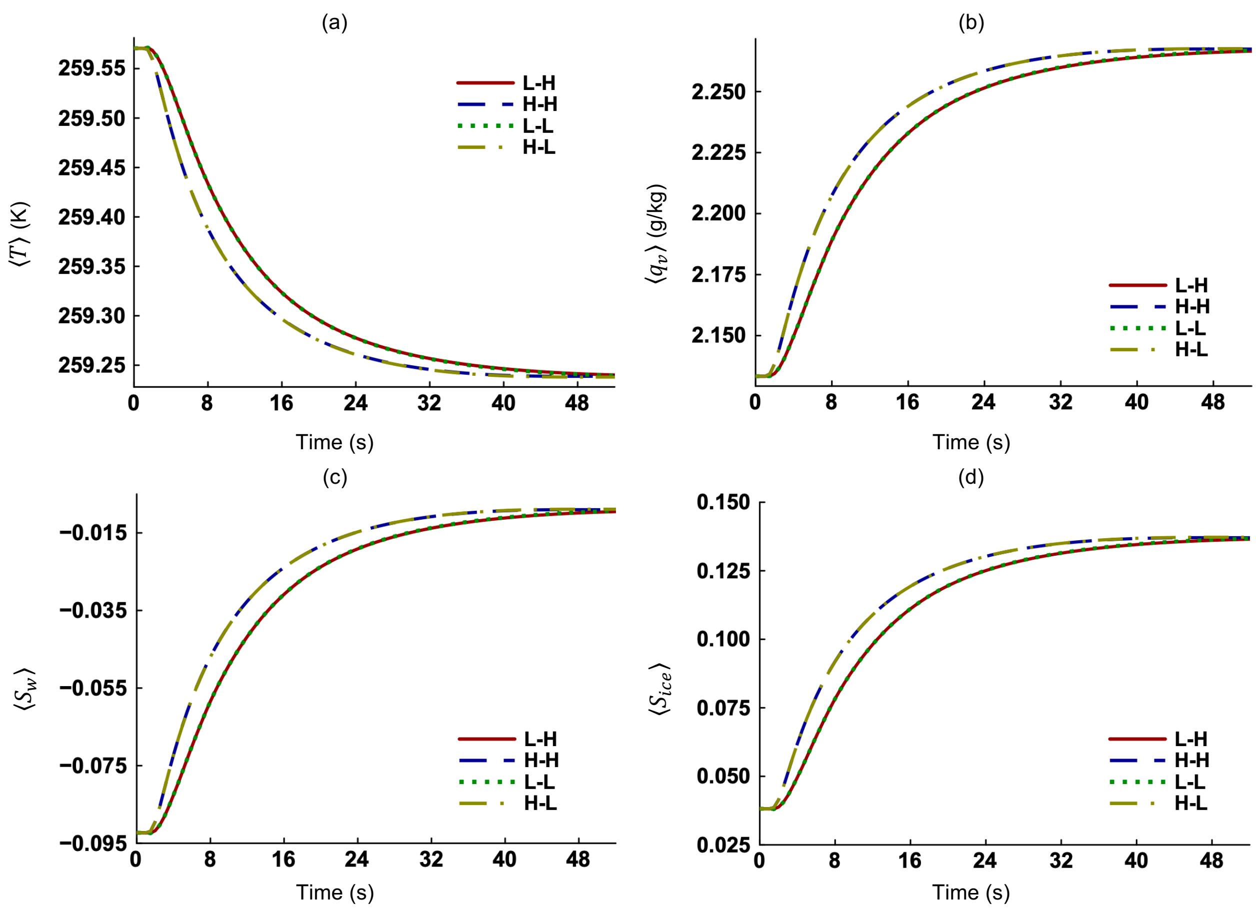


Figure 1. Time evolution of the (a) mean temperature ($\langle T \rangle$), (b) mean water vapor mixing ratio $\langle q_v \rangle$, (c) mean environmental supersaturation with respect to water ($\langle S_{e,w} \rangle$) and (d) mean environmental supersaturation with respect to ice ($\langle S_{e,ice} \rangle$).

Consistent with Figure 1(b), stronger evaporation for Cases H-H and H-L reduces the mean cloud droplet radius ($\langle r_w \rangle$) compared to the other two cases (Figure 2(a)). The mean radius of ice crystals ($\langle r_{ice} \rangle$) increases almost linearly for the four cases (Figure 2(b)). This observation agrees with the finding of Chen et al. (2024). Evaporation of cloud droplets produces more water vapor than ice growth can consume. Due to the availability of water vapor, ice crystals keep growing at the same pace at early and late times. However, evaporation slows down at late times as the $\langle S_{e,w} \rangle$ increases and the subsaturation (negative $S_{e,w}$) decreases. It appears that $\langle r_{ice} \rangle$ increases slightly faster for Case H-L compared to Case L-L, indicating that high turbulence has an expediting effect on ice growth. The same is true for Cases H-H and L-H. The masses of ice crystals are initially equal for the four cases, and they increase as expected during glaciation (Figure 2(c)). The increase is more significant when the ice mass is distributed among many but smaller ice crystals (Cases H-H and L-H) compared to few but larger ice crystals (Cases H-L and L-L). This is because more ice crystals translate into a larger surface area to which water vapor can deposit (Korolev & Issac, 2003). We also find that the distribution of ice mass in terms of ice crystal size and number density has more influence on ice growth than flow turbulence does. The ice mass fraction, which is the ratio of ice mass and total mass of liquid and ice, increases most for Case H-H, followed by Cases L-H, H-L, and L-L, in order (Figure 2(d)). Thus, a higher level of flow turbulence increases the rate of glaciation for any ice number density. Again, glaciation is expedited when ice crystals are smaller but more in number compared to the opposite for any flow turbulence level.

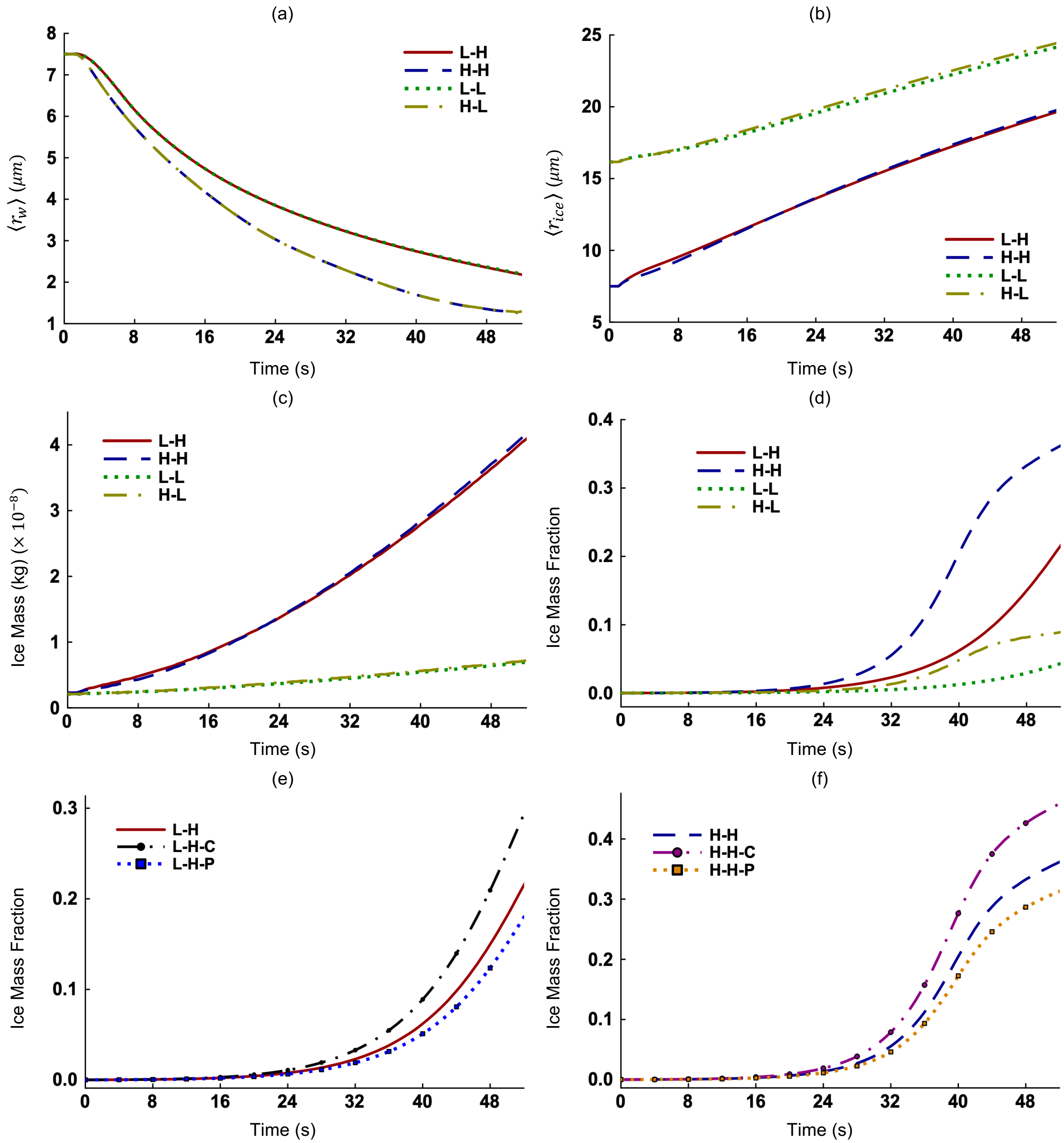


Figure 2. Time evolution of the (a) mean radius of cloud droplets ($\langle r_w \rangle$), (b) mean radius of ice crystals ($\langle r_{ice} \rangle$), (c) ice mass and (d) ice mass fraction, for Cases L-H, H-H, L-L, and H-L. Time evolution of the ice mass fraction for (e) Cases L-H, L-H-C, and L-H-P, and (f) Cases H-H, H-H-C, and H-H-P.

The aspect ratios (A) are 2.5 and 0.4 for ice crystals resembling hexagonal column and hexagonal plate, respectively, and they remain constant throughout the simulation time. However, the f, b, and h evolve dynamically with time. We see in Figures 2(e)-(f) that the ice mass fraction increases according to the order: Cases L-H-C > L-H > L-H-P and Cases H-H-C > H-H > H-H-P. Hence, ice mass growth and glaciation rate are fastest if ice crystals resemble hexagonal column and slowest if they are like hexagonal plate in our model. This holds true for both turbulence levels. Using molecular dynamics, Rozmanov & Kusalik (2012), Seo et al. (2012), and Shi et al. (2025) reported that the basal faces of hexagonal ice crystals grow slower than the prism faces. This is because basal faces grow layer by layer and vapor molecules need to complete a full rearrangement before the next layer begins to grow. The prism faces grow through a collective three-dimensional mechanism, which can incorporate vapor molecules readily. Since basal faces account for a larger proportion of the total surface area in a hexagonal plate than in a hexagonal column, ice growth is faster for the latter. Note that Westbrook et al. (2008) also used a molecular dynamics approach to determine the capacitance of hexagonal ice crystals. Hence, the agreement between our results and those of Seo et al. (2012), and Shi et al. (2025) is desirable. We also find that the growth rate of a spherical ice crystal is slower than that of a hexagonal column but faster than that of a hexagonal plate. Moreover, ice crystal shape has no impact on cloud evaporation and influences the glaciation process by affecting ice growth only.

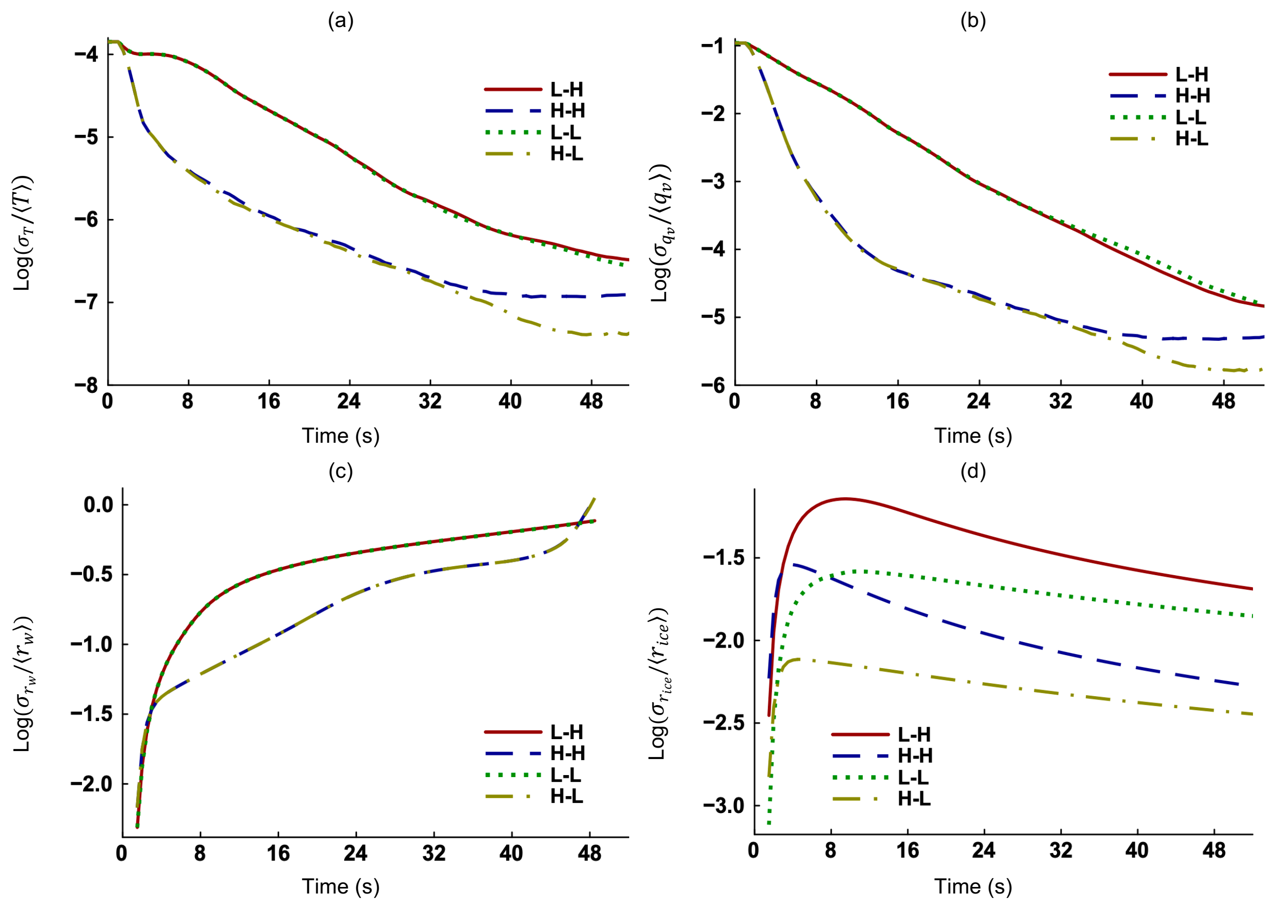
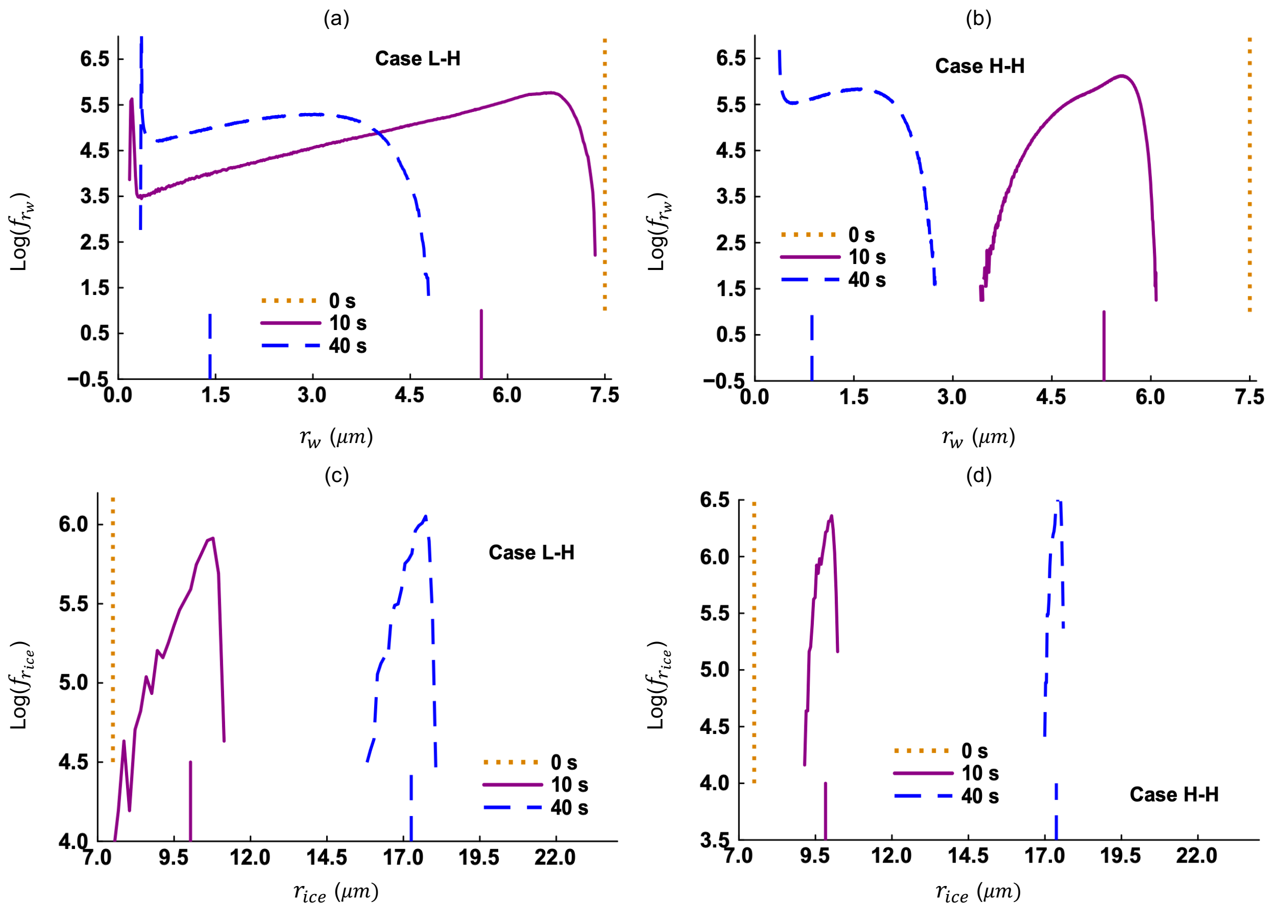


Figure 3. Time evolution of the relative dispersions of (a) temperature ($\sigma_T/\langle T\rangle$), (b) vapor mixing ratio ($\sigma_{q_v}/\langle q_v\rangle$), (c) cloud droplet radii ($\sigma_{r_w}/\langle r_w\rangle$), (d) ice crystal radii ($\sigma_{r_{ice}}/\langle r_{ice}\rangle$).

The time evolution of the relative dispersion, which is the standard deviation divided by the mean, is plotted in Figures 3(a)-(b) for temperature and vapor mixing ratio, respectively. We see that both $\sigma_T/\langle T\rangle$ and $\sigma_{q_v}/\langle q_v\rangle$ decrease with time because thermodynamic fields become more homogenized through turbulent mixing. The higher the turbulence level (Cases H-L and H-H), the quicker the homogenization and reduction of fluctuations (Sharfuddin et al., 2025). Figures 3(c)-(d) are analogous to Figures 3(a)-(b) but for cloud droplet and ice crystal radii, respectively. At early times, monodisperse cloud droplets and ice crystals are exposed to turbulent fluctuations, and their relative dispersions increase. The increase of $\sigma_{r_w}/\langle r_w \rangle$ slows down at late times (Figure 3(c)) and the $\sigma_{r_{ice}}/\langle r_{ice} \rangle$ decreases after reaching a peak (Figure 3(d)) for all four cases. We note that high flow turbulence has a retarding effect on the relative dispersion of ice crystals for any ice number density (Cases H-L vs L-L and H-H vs L-H). For similar levels of turbulence, high ice number density has an expediting effect on the $\sigma_{r_{ice}}/\langle r_{ice} \rangle$. The relative dispersion of cloud droplet radii is not affected by ice number density, but higher flow turbulence tends to retard the $\sigma_{r_w}/\langle r_w \rangle$. These results show that turbulent fluctuations influence T, q_v, r_w, and r_{ice} and thus glaciation.



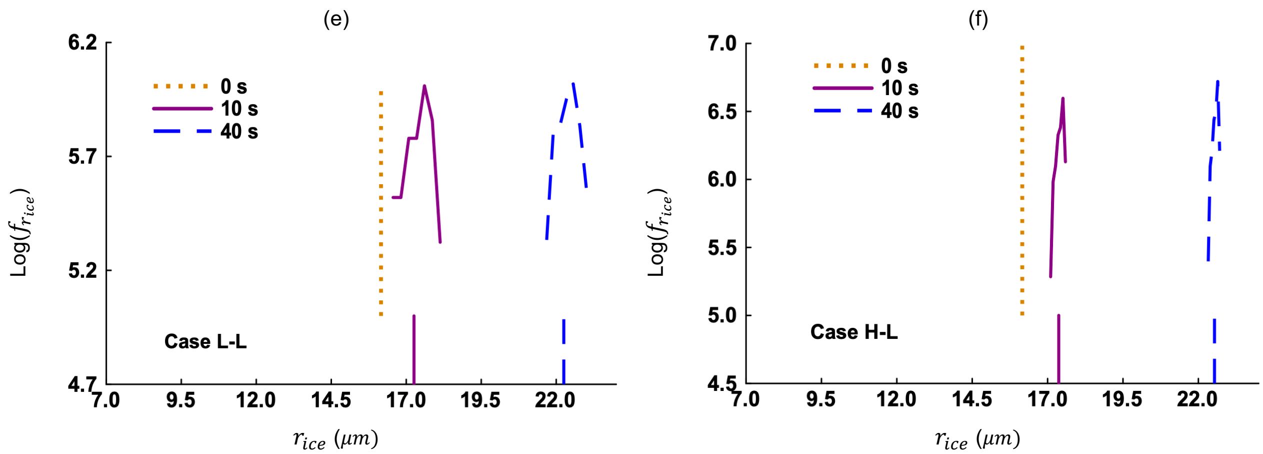


Figure 4. Log PDFs of cloud droplet radii (f_{r_w}) for Cases (a) L-H and (b) H-H. Log PDFs of ice crystal radii ($f_{r_{ice}}$) for Cases (c) L-H, (d) H-H, (e) L-L, and (f) H-L. Each plot shows the PDFs at 0 s, 10 s, and 40 s. The vertical line segments (bottom) indicate the mean values.

We examine the probability density functions (PDFs) of evaporating cloud droplets and growing ice crystals. The PDFs of cloud droplet radii (f_{r_w}) are shown in Figures 4(a)-(b) and those of ice crystal radii ($f_{r_{ice}}$) are shown in Figures 4(c)-(f). Both cloud droplets and ice crystals have monodisperse size (7.5 μm) distributions at 0 s. The f_{r_w} broadens between 0 s and 10 s for Case L-H because of turbulent fluctuations. It narrows between 10 s and 40 s as droplets experience partial evaporation and the maximum droplet radius decreases (Figure 4(a)). Also, the f_{r_w} appears to be bimodal for this case, with one peak for very small cloud droplets and another for larger droplets. The f_{r_w} are narrower for Case H-H (Figure 4(b)) at 10 s and 40 s than those for Case L-H due to enhanced mixing and reduced fluctuations (Figure 3(c)). The $f_{r_{ice}}$ is skewed to the left at 10 s and 40 s, and most ice crystals are larger than the mean for Case L-H (Figure 4(c)). It is found that the corresponding $f_{r_{ice}}$ are narrower when turbulence is stronger (Figure 4(d)) or ice number density is lower (Figures 4(e)-(f)). We see that some ice crystals have hardly grown from their initial 7.5 μm size by 10 s for Case L-H (Figure 4(c)). Whereas no ice crystal is smaller than 9 μm for Case H-H at 10 s (Figure 4(d)), although the $\langle r_{ice} \rangle$ is smaller for this case. This demonstrates the impact of turbulent mixing on glaciation.

5 Conclusions

We have studied the glaciation of turbulent mixed-phase clouds using a three-dimensional DNS model with Lagrangian microphysics. Our results clarify several microphysical aspects of glaciation. A high level of flow turbulence intensifies the mixing between cloud and environment and expedites glaciation. Glaciation is also enhanced if the ice mass is distributed among smaller but many ice crystals compared to the opposite. A hexagonal column shape promotes ice growth and glaciation while a hexagonal plate geometry does the opposite. These findings are the key to accurate estimation of the mass fractions of cloud droplets and ice crystals that co-exist in mixed-phase clouds. The present letter also shows that particle-based DNS has become an important tool to explicitly resolve ice microphysics. We shall employ this approach for future investigation of more complex ice crystal geometries and ice growth processes, with consequences for glaciation.

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Open Research

The data that support the findings of this work are available from the corresponding author upon reasonable request.

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